

A Python Package for the Simultaneous Global Sensitivity Analysis of Foreground and Background Parameters in Life Cycle Assessment



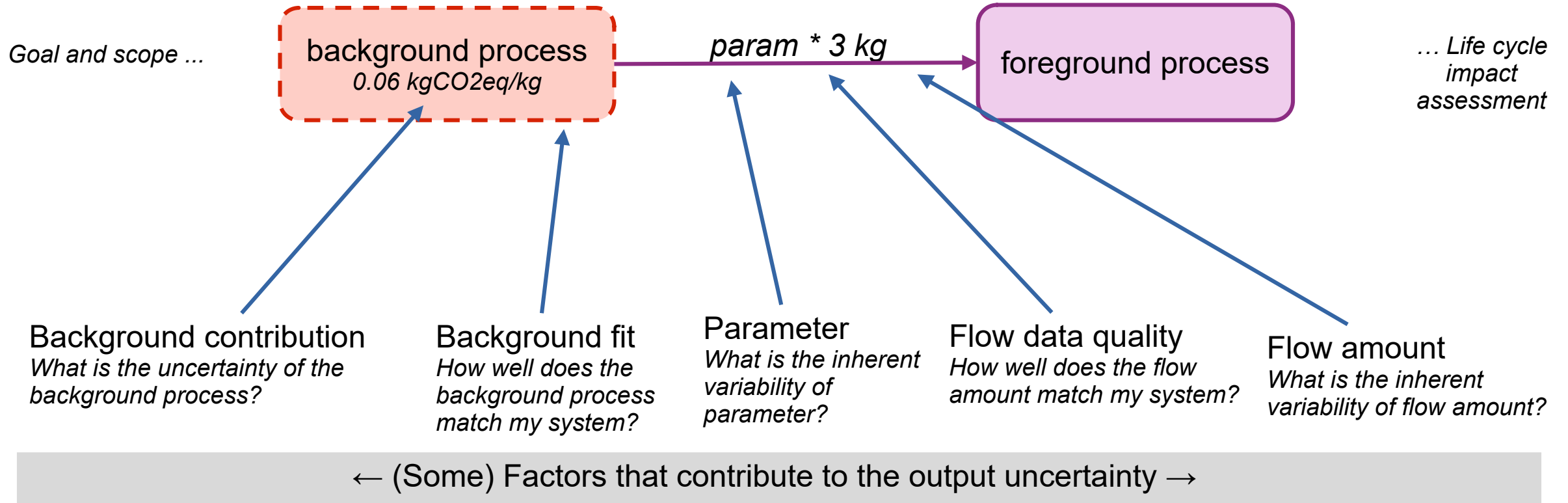
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Deciding between two alternatives

Method	No uncertainty analysis	Monte Carlo simulation	Discernibility analysis
Alternative A	1.1 kg CO ₂ eq	1.1 ± 0.1 kg CO ₂ eq	A is lower than B in 60% of cases
Alternative B	1.2 kg CO ₂ eq	1.2 ± 0.2 kg CO ₂ eq	
Conclusion	A is lower	Not clear	Not clear
	Bad	Good	Better

How to increase the discernibility?
→ Reduce the uncertainty in the model!

Uncertainty in LCA Models



Uncertainty Apportioning

Background contribution
What is the uncertainty of the background process?

Results from 1000s of underlying parameters

Existing implementations either
1) ignore background, or
2) consider all 1000s of background parameters.

Two categories of methods:

Random sampling

Sobol (Li & Mahadevan estimator)
Li and Mahadevan, 2016

XGB Feature Importance
Chen, 2016

Spearman rank correlation
Spearman, 1904

Standardized regression coefficients
Xu and Gertner, 2008

TreeSHAP
Lundberg, 2019

PAWN
Pianosi and Wagener, 2018

RBD-FAST
Tarantola et al., 2006

Delta
Plischke et al., 2013

Specific sampling

Sobol (Saltelli sampling)
Saltelli, 2007

FAST
Saltelli et al., 1999

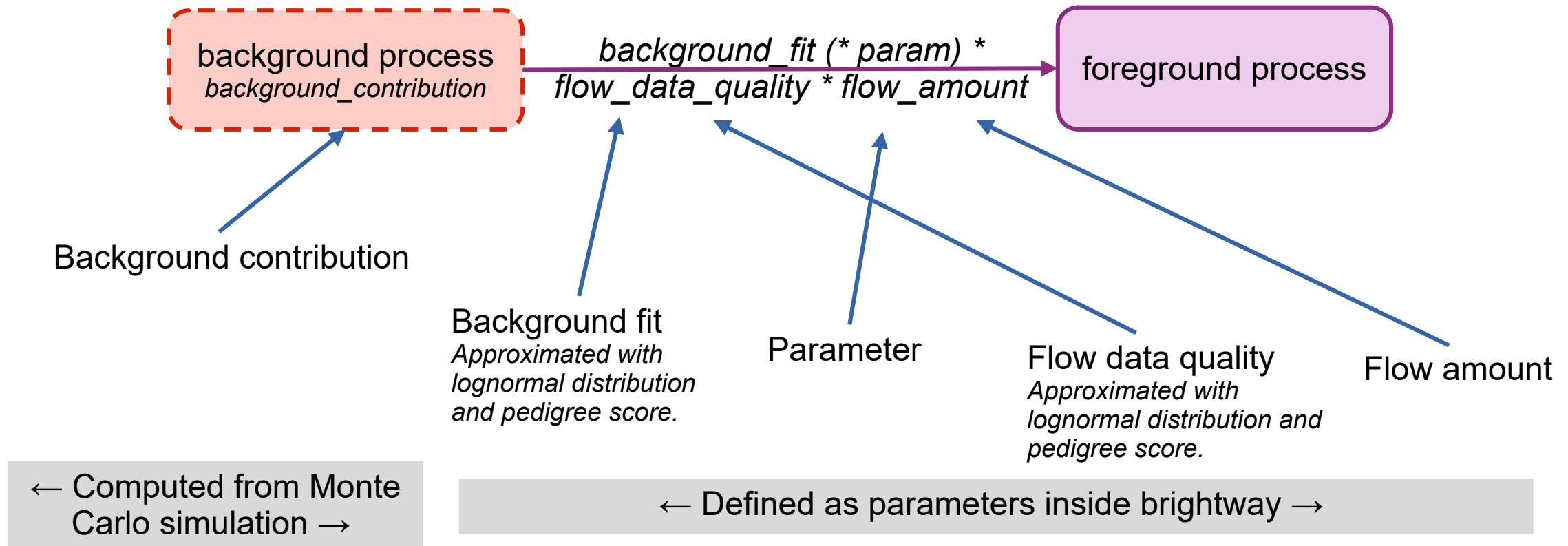
Fractional factorial
Saltelli, 2007

Method of Morris
Campolongo et al., 2007

Existing (Free) Solutions

Name	Flow amount	Flow data quality	Parameter	Background fit	Background contribution	Uncertainty apportioning
<i>Software</i>						
openLCA	✓	✓	✓	●	● * not shown for individual Monte Carlo iterations	●
brightway (Python)	✓	✓ * can be manually added as flow uncertainty	● * no uncertainty	●	✓	●
gsa_framework (Python)	✓	●	●	●	✓	✓
lca_algebraic (Python)	✓	✓	✓	●	✓	✓
...	Did we forget your solution? 🤖 Let us know!					
lcatoolbox (Python)	✓	✓	✓	✓	✓	✓
<i>Papers</i>						
Kim et al., 2025	✓	●	●	●	✓	✓
Kaddoura et al., 2025	●	✓	✓	✓	✓	✓

Our Package: Icatoolbox



Our Package: Icatoolbox



Features

- Import LCA model from spreadsheet
- Apply corrections to uncertainty definitions in ecoinvent
- Contributions analysis
- Uncertainty analysis via Monte Carlo simulation
- Discernibility analysis
- Local sensitivity analysis
- Uncertainty apportioning
 - Supports Sobol (from Saltelli or Monte Carlo), FAST, RBD-FAST, PAWN, Delta, Spearman, Gradient Boosting, Linear Regression.
 - Aggregate results based on discernibility of individual impact categories

Scope

Satisfy the needs of our research group.

Future developments

Depend on who is interested...

Input: LCA model in a spreadsheet

All sheet beginning with „a_“ define new activities.

	A	B	C	D	E	K	L	M	N	O	P	Q	R
1	Name	PCB assembly, power+sensor [common]											
2	Location	DE											
3	Product	assembled PCB, power+sensor [common]											
4	Amount	1											
5	Unit	item(s)											
6	Notes												
7													
8	Description	Group	Amount	Unit	Uncertainty Type	Activity	Product	Type	Location	Categories	Data Quality	Activity Fit	Note
9	1x INA234 (1)		ic_ina234_di	kilogram	No uncertainty	market for integrated ci	technospl		GLO		(1;1;1;1)	(1;1;1;4)	
10	13x Capacitor, SMD, 0603 (C1, C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, C12, C13)		13*cap_smd_	kilogram	No uncertainty	market for capacitor, fo	technospl		GLO		(1;1;1;1)	(1;1;1;2)	
11	2x SIP32431DR3-T1G3 (IC1, IC2)		2*ic_sip3243	kilogram	No uncertainty	market for integrated ci	technospl		GLO		(1;1;1;1)	(1;1;1;4)	
12	1x inductor, SMD, 2813 (L1)		1*ind_smd_2	kilogram	No uncertainty	market for inductor, low	technospl		GLO		(1;1;1;1)	(1;1;1;2)	
13	1x inductor, SMD, 3816 (L2)		1*ind_smd_3	kilogram	No uncertainty	market for inductor, low	technospl		GLO		(1;1;1;1)	(1;1;1;2)	
14	1x resistor, SMD, 0805 (R1)		1*res_smd_0	kilogram	No uncertainty	market for resistor, surf	technospl		GLO		(1;1;1;1)	(1;1;1;2)	
15	11x resistor, SMD, 0603 (R2, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13)		11*res_smd_1	kilogram	No uncertainty	market for resistor, surf	technospl		GLO		(1;1;1;1)	(1;1;1;2)	
16	1x resistor, SMD, 2512 (R3)		1*res_smd_2	kilogram	No uncertainty	market for resistor, surf	technospl		GLO		(1;1;1;1)	(1;1;1;2)	

Input: LCA model in a spreadsheet

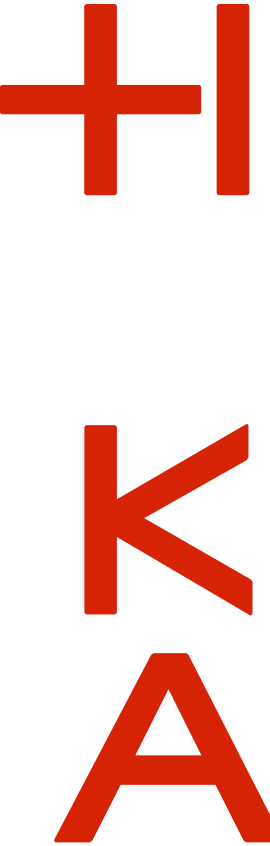
All sheet beginning with „p_“ define new parameters.

	A	B	C	D	E	F	G	H	
1	Name	Value	Uncertainty Type	Uncertainty Location	Uncertainty Scale	Uncertainty Shape	Uncertainty Minimum	Uncertainty Maximum	Note
2	ic_ina234_die_area_mm2	0,4044456	Triangular	0,4044456			0,1011114	1,0223486	BGA, 1.12346 mm ² package, die is
3	ic_sip32432_die_area_mm2	0,6174	Triangular	0,6174			0,1323	1,4553	SOP, 2.1x2.1mm ² , die is 0.03 – 0.3
4	ic_as7341_die_area_mm2	1,736	Triangular	1,736			0,31	3,1	DFN, 2.0x3.1mm ² , die is 0.05-0.5
5	ic_shtc3_die_area_mm2	1,12	Triangular	1,12			0,2	2	DFN, 4mm ² , die is 0.05-0.5 of hou
6	ic_tpsm83100s_die_area_mm2	1,456	Triangular	1,456			0,26	2,6	DFN, 2.0x2.6mm ² , die is 0.05-0.5
7	ic_bq25570_die_area_mm2	2,7083	Triangular	2,7083			0,666125	6,66125	QFN, 3.65x3.65mm ² , die is 0.05-0
8	ic_msp430fr5969_die_area_mm2	13,72	Triangular	13,72			2,45	24,5	QFN, 7x7mm ² , die is 0.05-0.5 of b
9	ic_die_to_mass_ratio_cm2_per_kg	180	Triangular	180			125	1000	Thesis from Paul Teehan, https://d
10	cap_smd_0603_mass_mg	6	Uniform				3	9,1	muRata MLCC Table, https://www
11	cap_smd_0805_mass_mg	15	Uniform				6,4	23	muRata MLCC Table, https://www
12	cap_tht_35_8_mass_g	1,9	Triangular	1,9			1,7	2,1	VEL08203R8306G Datasheet, http
13	ind_smd_3816_mass_mg	103	Triangular	103			51,5	154,5	https://www.mouser.sg/en/Produ
14	ind_smd_2813_mass_mg	32	Triangular	32			16	48	https://www.mouser.lu/de/Produ
15	ind_smd_0805_mass_mg	10	Triangular	10			5	15	https://www.tdk-electronics.tdk.c
16	res_smd_0805_mass_mg	5	Triangular	5			2,5	7,5	stackpole RCMF, https://www.sei
17	res_smd_0603_mass_mg	1,9	Triangular	1,9			0,95	2,85	stackpole RCMF, https://www.sei

◀ ◁ ▷ ▶ + p_parameters a_all_phases_B | a_all_phases_S | a_prod | a_prod_B | a_prod_S | a_prod_pcb_power_sensor | a_prod_pcb_power_sensor_B | a_prod_pcb_power_sensor_S | a_prod_pcb_mcu_lora

Time for a Demonstration!





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- Have a look at the package: https://github.com/HKA-IES/lca_toolbox
- Stay tuned for upcoming paper:
 - How to aggregate uncertainty apportioning results in the case of multiple outputs?
 - Which uncertainty apportioning method ranks converge faster?
 - Case study of an environmental sensor (solar- vs battery-power)

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- F. Campolongo, J. Cariboni, and A. Saltelli, “An effective screening design for sensitivity analysis of large models,” *Environmental Modelling & Software*, vol. 22, no. 10, pp. 1509–1518, Oct. 2007, doi: 10.1016/j.envsoft.2006.10.004.
- T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, in KDD ’16. New York, NY, USA: Association for Computing Machinery, Aug. 2016, pp. 785–794. doi: 10.1145/2939672.2939785.
- M. Kaddoura, G. Majeau-Bettez, B. Amor, and M. Margni, “Global sensitivity analysis reduces data collection efforts in LCA: A comparison between two additive manufacturing technologies,” *Science of The Total Environment*, vol. 975, p. 179269, May 2025, doi: 10.1016/j.scitotenv.2025.179269.
- A. Kim, C. Mutel, and S. Hellweg, “Global sensitivity analysis of correlated uncertainties in life cycle assessment,” *Journal of Industrial Ecology*, vol. 29, no. 4, pp. 1090–1104, 2025, doi: 10.1111/jiec.70036.
- C. Li and S. Mahadevan, “An efficient modularized sample-based method to estimate the first-order Sobol’ index,” *Reliability Engineering & System Safety*, vol. 153, pp. 110–121, Sep. 2016, doi: 10.1016/j.ress.2016.04.012.
- S. M. Lundberg, G. G. Erion, and S.-I. Lee, “Consistent Individualized Feature Attribution for Tree Ensembles.” *arXiv*, Mar. 2019. doi: 10.48550/arXiv.1802.03888.
- F. Pianosi and T. Wagener, “Distribution-based sensitivity analysis from a generic input-output sample,” *Environmental Modelling & Software*, vol. 108, pp. 197–207, Oct. 2018, doi: 10.1016/j.envsoft.2018.07.019.
- E. Plischke, E. Borgonovo, and C. L. Smith, “Global sensitivity measures from given data,” *European Journal of Operational Research*, vol. 226, no. 3, pp. 536–550, May 2013, doi: 10.1016/j.ejor.2012.11.047.
- A. Saltelli, S. Tarantola, and K. P.-S. Chan, “A Quantitative Model-Independent Method for Global Sensitivity Analysis of Model Output,” *Technometrics*, 1999.
- A. Saltelli, Ed., *Global sensitivity analysis: The primer*. Chichester, England ; Hoboken, NJ: John Wiley, 2007.
- C. Spearman, “The Proof and Measurement of Association between Two Things,” *The American Journal of Psychology*, vol. 15, no. 1, pp. 72–101, 1904, doi: 10.2307/1412159.
- S. Tarantola, D. Gatelli, and T. A. Mara, “Random balance designs for the estimation of first order global sensitivity indices,” *Reliability Engineering & System Safety*, vol. 91, no. 6, pp. 717–727, Jun. 2006, doi: 10.1016/j.ress.2005.06.003.
- C. Xu and G. Z. Gertner, “Uncertainty and sensitivity analysis for models with correlated parameters,” *Reliability Engineering & System Safety*, vol. 93, no. 10, pp. 1563–1573, Oct. 2008, doi: 10.1016/j.ress.2007.06.003.